



Yeshiva University
SY SYMS SCHOOL OF BUSINESS
Academic Advisement



Business Analytics Course List (as of 10-1-2022)

- **IDS 1001 Business Algebra 3 credits**
This course provides a review of computational and problem-solving skills. Included is a presentation of a broad scope of fundamental mathematical concepts in applied mathematics relevant to accounting, finance, management, and marketing, with examples drawn from different business disciplines.
- **IDS 1010 Mathematics for Business**
The objectives of this course are to provide an overview of (i) algebra, (ii) functions including exponential and logarithmic functions, (iii) their application to business through the study of the time value of money, and an introduction to the application of calculus and optimization to business.
- **IDS 1020 Introduction to Information Systems 3 Credits**
This course provides the background necessary to make decisions about computer-based information systems and to be an end-user. The course includes hands-on experience with personal computers and information systems management. Groups and individual computer assignments expose students to electronic spreadsheet analysis and database management on a personal computer. Management aspects focus on understanding computer technology, systems analysis and design, and control of information processing by managers.
- **IDS 1131 Statistics for Business 3 Credits**
This course includes modern statistical methods as a basis for decision making. Topics include fundamentals of probability, discrete and continuous distributions, descriptive statistics, and inferential statistics. Credit is not given for STA 1021.
Pre-requisite: **IDS 1010** for students starting in Fall 2021. **IDS 1001** for students starting before Fall 2021.
- **IDS 1300 Business Calculus 3 Credits**
Applications of calculus for solving business problems. Topics include functions, limits, techniques of differentiation, marginal analysis, higher-order derivatives and optimization, integration, and applications of these techniques as they relate to business. Prerequisites: **IDS 1001, 1131**. Only available for students who started before Fall 2021.
- **IDS 1400 Regression Analysis 1 Credit**
Linear Regression Analysis: Topics include the simple linear regression model, inference in regression analysis, sensitivity analysis, multiple regression analysis, and introduction to time series analysis. Only available for students who started before Fall 2021.
- **IDS 1456 Quantitative Methods for Management 3 Credits**
Part 1. Applications of calculus for solving business problems. Topics include functions, limits, techniques of differentiation, marginal analysis, higher-order derivatives and optimization, integration, and applications of these techniques as they relate to business. Part 2: Linear Regression Analysis: Topics include the simple linear regression model, inference in regression analysis, sensitivity analysis, multiple regression analysis, and introduction to time series analysis.
Prerequisites: **IDS 1001, 1131**. Only available for students who started before Fall 2021.
- **IDS 1556 Regression Analysis and Visualization**
The objectives of this course are to provide an overview of simple linear regression, multiple regression, and logistic regression. The material is focused on examples relevant to business applications. In addition to learning the specifics of regression, students also learn how to generate effective visualizations as part of the modeling process and as part of the presentation and reporting process within a business context.
Prerequisites: **IDS 1010, IDS 1131**.
- **IDS 2020 Data Visualization**
Students learn how to generate effective data visualizations as well as critique and improve visualizations that have already been generated. Additionally, students learn what types of visualizations are appropriate when storytelling. Practically speaking, after successfully completing this course, students will be comfortable using

R/RStudio as well as Tableau for generating effective data visualizations.

Prerequisite: **IDS 1020**

- **IDS 2030 Business Analytics and Programming 3 credits**

Today, more than ever, businesses must learn to leverage their data as a strategic resource. This course introduces the tools and techniques used by data scientists, marketers, and analysts to understand, manipulate, and present the data that is the lifeblood of enterprises today. Students will learn modern techniques related to data manipulation, storage, retrieval, and computer programming.

Pre or Corequisites: **IDS 1131 and IDS 1020.**

- **IDS 2160 Decision Models 3 Credits**

This course introduces the basic principles and techniques of applied mathematical modeling via spreadsheets for managerial decision making. Students learn to use some of the more important analytic methods focusing on spreadsheet modeling. Students learn to develop models that can be used to improve decision making within an organization; sharpen their ability to structure problems and to perform logical analyses; translate descriptions of decision problems into formal models and investigate those models in an organized fashion; identify settings in which models can be used effectively and apply modeling concepts in practical situations. The course emphasizes model formulation and interpretation of results and is aimed at undergraduate students with little prior exposure to modeling and quantitative analysis, but it is appropriate for all students who wish to strengthen their spreadsheet and quantitative skills. The emphasis is on models that are widely used in diverse industries and functional areas, including finance, operations, and marketing.

Prerequisites: **FIN 1001, IDS 1020, IDS 1131, MAR 1001.**

- **IDS 2460 Data Management for Business Analytics**

This course surveys several approaches and tools for working with data. This includes tools to help the data scientist automate extracting the data from native data stores, manipulating, and converting the data to other formats and presenting data using effective visualizations.

Prerequisites: **IDS 2030.**

- **IDS 2550 Statistical Learning for Business Analytics 3 Credits (Same as ENT/MAR 2550)**

Statistical Learning and data mining are powerful tools that help companies focus on the most important information in the data they have collected and or purchased. about the behavior of their customers and potential customers. It discovers information within the data that queries, and reports can't effectively reveal. This course explains what data mining is, how it can be used, and how it can help a company leapfrog its competition. Internet based applications such as social media, website usage, tracking and online reviews as well as a firm's own activities and business processes, are discussed as potential sources of data.

Prerequisite: **IDS 1020, IDS 1131, and IDS 2030.**

- **IDS 3000 Business Analytics Capstone 3 Credits**

This capstone course focuses on the integration of various methods and technologies into data analytics professional encounter. Students are required to complete a project simulating a real-world data analytics environment. Specifically, students need to query databases, organize data, apply appropriate statistical models, utilize appropriate software and data science packages, and finally present their findings within a business context.

Prerequisites: **IDS 2030, IDS 2460, and IDS 2550.**

- **IDS 3800H Data Driven Decision Making**

The purpose of this course is to provide the student with the exposure to the big picture in data analytics. Although there will be some statistics, mathematics, and coding in the course, the focus will be on developing critical thinking approaches within the context of data analytics. In other words, although students will be responsible for performing some of the analytics, they will be graded on how they utilize the analytics performed to make recommendations as well as to what extent they have critically analyzed the big picture.

Prerequisites: **IDS 1131.**